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A Fractional Fuzzy Dynamical Model for Math-Phobia with Optimal Learning Intervention Strategies

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⁴ Hence is contraction. The Banach contraction principle guarantees that the suggested model (1) has just one solution.

⁵ Including these controls, the fuzzy fractional differential equations become:

⁶ According to the Pontryagin's maximum principle, define the Hamiltonian function as follows:

⁷ Figure 4. Trajectory of intervention-aware students (t) for distinct fractional orders ϑ .

Abstract: This paper introduces a new six-compartment fuzzy fractional-order epidemic model that describes the spread and mitigation of math-phobia in student populations. The susceptible $S(t)$, mathphobic $M(t)$, not interested $N(t)$, intervention-aware (t), interested (t), and recovered (t) compartments capture key behavioural transitions influenced by psychological, social, and educational factors. Caputo-type fuzzy fractional differential equations are used to derive these dynamics, considering memory effects and epistemic uncertainty in the transmission and recovery rates. By applying fixed-point theory, we prove the existence and uniqueness of the solution under fuzzy initial conditions, hence guaranteeing mathematical coherence of the model.